MSE 360: Kinetic Processes in Materials
Homework #10

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

A silicon wafer is being oxidized in an atmosphere of pure oxygen at 1200 K. The thickness of the oxide, x (in µm), as a function of time, t (in seconds) is given by the parabolic oxidation law:

$$x^2 + Bx = Ct$$

where $B = 0.040$ µm and $C = 0.045$ µm^2/s.

a) On a separate sheet, make a computer-drawn plot of the oxide thickness x as a function of time t for oxide thicknesses between 0 and 20 µm.

b) What is the rate of change of the oxide thickness (um/s) when the oxide is 10 µm thick?

c) When the oxide is 10 µm thick, is the oxidation reaction controlled by diffusion through the oxide film or by the reaction at the Si/SiO_2 interface? Justify your answer.

d) If air is used in the reaction chamber instead of pure oxygen, do you expect the oxidation rate to change when $x = 10$ µm? If so, will it increase or decrease? Fully explain your answer.

e) If the temperature in the reaction chamber is decreased to 1000 K, do you expect the oxidation rate to change when $x = 10$ µm? If so, will it increase or decrease? Fully explain your answer.